Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: compatibility with $(\boldsymbol{P}_\Gamma)$. The natural pullback, followed by forgetting volume forms and passing to the full unmeasured tensor ring, does land in the unmeasured tensor kernel. This checks the required compatibility with the RES retraction.

Write $q = \mu\Psi^{\mathrm{db}} : T_\mu^{\mathrm{db}} \longrightarrow B = \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*]$ and $\Psi : T \longrightarrow A = \mathbf{K}\,\mathrm{RV}[*]$. Let $f : T_\mu^{\mathrm{db}} \longrightarrow T$ and $u : B \longrightarrow A$ forget the forms and include bounded objects into the full categories. The product-lift definitions give

$$\Psi \circ f = u \circ q. \qquad J = (\boldsymbol{P}_\Gamma) \subseteq B, \qquad K = \ker \Psi.$$

The precise inclusion is

$$\boxed{f\big(q^{-1}(J)\big) \subseteq K.}$$

Compute the annular generators. For every generator with $\gamma \geq 0$ and the stipulated original-ground definable $t_\gamma \in \gamma^\sharp$, the annulus is the full lift of $(0, \gamma]$, with its right endpoint included. Put $H_a = (a, \infty)$. In the full unmeasured ring,

$$\begin{aligned} u(\boldsymbol{P}_\gamma) &= [(0,\gamma]^\sharp] + [\{t_\gamma\}]_1 - [\{1\}]_1 \\ &= [H_0^\sharp] - [H_\gamma^\sharp] = 0. \end{aligned}$$

The singleton classes agree, and multiplication by t_γ is an original-ground RV bijection between the two tails. This is a ring calculation, using additive group cancellation. It neither inverts a torus class nor adjoins a new RV parameter. At $\gamma = 0$ the annulus is empty and the singleton terms cancel as well.

Since u is a ring homomorphism, the calculation proves $u(J) = 0$ for the whole ideal. For any $z \in q^{-1}(J)$, including every possible tensor-preimage ambiguity,

$$\Psi(f(z)) = u(q(z)) = 0.$$

This proves the boxed inclusion without needing the full description of $\ker \Psi$ or injectivity of any bounded-to-unbounded map.

Consequence for the RES evaluation. The balanced evaluation $\widetilde{\epsilon}_b : T \longrightarrow !\mathbf{K}\,\mathrm{RES}[*]$ annihilates K, hence

$$(\widetilde{\epsilon}_b \circ f)\big(q^{-1}(J)\big) = 0.$$

Thus the retraction descends through $(\boldsymbol{P}_\Gamma)$ with no further RES quotient beyond $!I[*]$. Directly on a canonical lift the value is $\chi((0,\gamma])[\mathbb{G}_m]_1 + [1]_1 - [1]_1 = 0$. The subsequent forgetting of the grading gives $\mathbb{E}^\diamond$. The tail cancellation above takes place in the *full* unmeasured ring, and is not asserted within the bounded ring.

Retain both ideals in the bounded tensor presentation. Let I_{ann} be generated by the canonical lifts

$$\widetilde{P}_\gamma = 1 \otimes [((0,\gamma],0)]_1 + [(\{t_\gamma\},0)]_1 \otimes 1 - [(\{1\},0)]_1 \otimes 1.$$

Each satisfies $f(\widetilde{P}_\gamma) = 0$ already in T. Surjectivity gives the correct quotient presentation

$$q^{-1}(J) = \ker q + I_{\mathrm{ann}}, \qquad B/J \cong T_\mu^{\mathrm{db}}/(\ker q + I_{\mathrm{ann}}).$$

The tensor kernel cannot in general be omitted from this sum. Over $\mathbb{S} = \mathbb{C}((t))$, take $R = \{r : r^5 = \mathrm{rv}(t)\}$, $\delta = 1/5$, $U = (0,\delta]$, $V = (\delta, 2\delta]$. Then

$$k = [(R,0)]_0 \otimes \big([(U,-\mathrm{id})]_1 - [(V,-\mathrm{id})]_1\big) \in \ker q \setminus I_{\mathrm{ann}}.$$

The root multiplication bijection preserves the zero combined weights. After f, the earlier endpoint detector sends k to $5(-1-1)\varepsilon = -10\varepsilon \neq 0$, whereas it kills I_{ann} because $f(I_{\mathrm{ann}}) = 0$. This non-absorption is different from the boxed inclusion, which holds; both ideals are annihilated by the RES evaluation.

The coefficient issue matters for this same pullback: replace R in k by the earlier genus-two RES[1] curve X, with form zero. The resulting bounded measured kernel element lies in $q^{-1}(J)$. After f, the curve detector gives $-2\eta\varepsilon \neq 0$ and annihilates the finite-coefficient ideal. Thus a finite-only version of K would fail this check, while the full all-X ideal passes it.

The detailed pullback, endpoint, and quotient checks are recorded in the section's P_Γ verification note. This settles the stated RES-retraction compatibility, not every later assertion in the paper. These calculations are mathematical proofs rather than completed concrete Lean exports. No HK comparison is assumed, and the approved working manuscript remains unchanged.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of $\mathrm{RV}[*]$ then it may also denote the object $(\boldsymbol{U}, 0)$ of $\mu\mathrm{RV}[*]$ with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+\,\mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+\,\mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+\,\mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}^{\circ\circ}_\infty] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+\,\mathrm{RV}[\leq 1]$.

Let I_{sp} be the (nonhomogeneous) semiring congruence relation on $\mathbf{K}_+\,\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}_\infty])$. Let

$$\boldsymbol{P} = [1] - [\mathrm{RV}^{\circ\circ}] \in \mathbf{K}\,\mathrm{RV}[1].$$

The corresponding principal ideal of $\mathbf{K}\,\mathrm{RV}[*]$ is thus generated by the element $\boldsymbol{P} - 1$.

Similarly, let $\mu\mathrm{I}_{\mathrm{sp}}$ be the semiring congruence relation on $\mathbf{K}_+\,\mu\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}])$, which is homogeneous, and the corresponding principal ideal of $\mathbf{K}\,\mu\mathrm{RV}[*]$ is generated by the element $\boldsymbol{P}$.

Resolved referee R051 [Typo] – Spelling of homogeneous.
Original source line 1358. In this notation, replace "nonhomogenous" and "homogenous" by "nonhomogeneous" and "homogeneous," respectively.

Author resolution. The author approves correcting the spellings to nonhomogeneous and homogeneous. The congruences and their generators are unchanged.

Notation 3.21. For each $\boldsymbol{U} = (U, f) \in \mathrm{RV}[k]$, let U_f be the set $\bigcup\{f(u)^\sharp \times u \mid u \in U\}$. Let $\mathbb{L}_{\leq k} : \mathrm{RV}[\leq k] \longrightarrow \mathrm{VF}[k]$ be the map given by $\boldsymbol{U} \longmapsto U_f$. Set $\mathbb{L} = \bigcup_k \mathbb{L}_{\leq k}$.

Let $\mu\mathbb{L}_k : \mu\mathrm{RV}[k] \longrightarrow \mu\mathrm{VF}[k]$ be the map given by $(\boldsymbol{U}, \omega) \longmapsto (\mathbb{L}\boldsymbol{U}, \mathbb{L}\omega)$, where $\mathbb{L}\omega$ is the obvious function on $\mathbb{L}\boldsymbol{U}$ induced by ω. Set $\mu\mathbb{L} = \bigoplus_k \mu\mathbb{L}_k$.

Suppose that $\mathbb{S}$ is (VF, Γ)-generated. The map $\mathbb{L}$ induces a surjective semiring homomorphism $\mathbf{K}_+\,\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mathrm{VF}_*$ and the map $\mu\mathbb{L}$ induces a surjective graded semiring homomorphism $\mathbf{K}_+\,\mu\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{VF}[*]$, see [8, § 4, § 6] or [17, Corollaries 7.7, 10.6]; we use the same notation for these homomorphisms as well as their groupifications. We have

Resolved referee R053 [Error; revised] – Scope of the parameter hypothesis.
Original source line 1366. State here the (VF, Γ)-generation hypothesis used in Theorem 3.22, before asserting that lifting induces these homomorphisms. Without a parameter restriction, even well-definedness can fail. For example, let the parameters be generated by a trivially valued algebraically closed field C and a named $\tau \in \mathrm{RV}$ of positive value. The singleton objects $\{\tau\}$ and $\{1\}$ are isomorphic in $\mathrm{RV}[1]$. But $\tau^\sharp$ has no definable VF point: Lemma 2.13 places every such point in $\mathrm{acl}(C) \cap \mathrm{VF} = C$. In contrast, $1 + \mathcal{M}$ contains the definable point 1, so their lifts cannot be definably bijective. Moving the stated hypothesis forward removes this obstruction.

Author resolution. The author approves stating the (VF, Γ)-generation hypothesis before both homomorphism assertions. The lift on individual objects needs no such hypothesis. Under the standing nontrivial-ground-value-group assumption, take instead a small algebraically closed nontrivially valued subfield F and name $\tau \in \mathrm{RV}$ with $\mathrm{vrv}(\tau) \notin \Gamma(F)$. Lemma 2.13 puts every definable VF point over the